HPC-Ops

Tencent Hunyuan's production operator library for LLM inference

Tencent Hunyuan AI Infra · HPC-Ops

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Frameworks & Platform Stacks · github.com/Tencent/hpc-ops

Tencent serves its Hunyuan models on HPC-Ops

- CUDA operator library from Tencent Hunyuan AI Infra
- Tuned for NVIDIA **H20** (SM90); runs Tencent's large-scale serving
- Open source since Jan 2026 (MIT); in **SGLang** main since Aug 2026

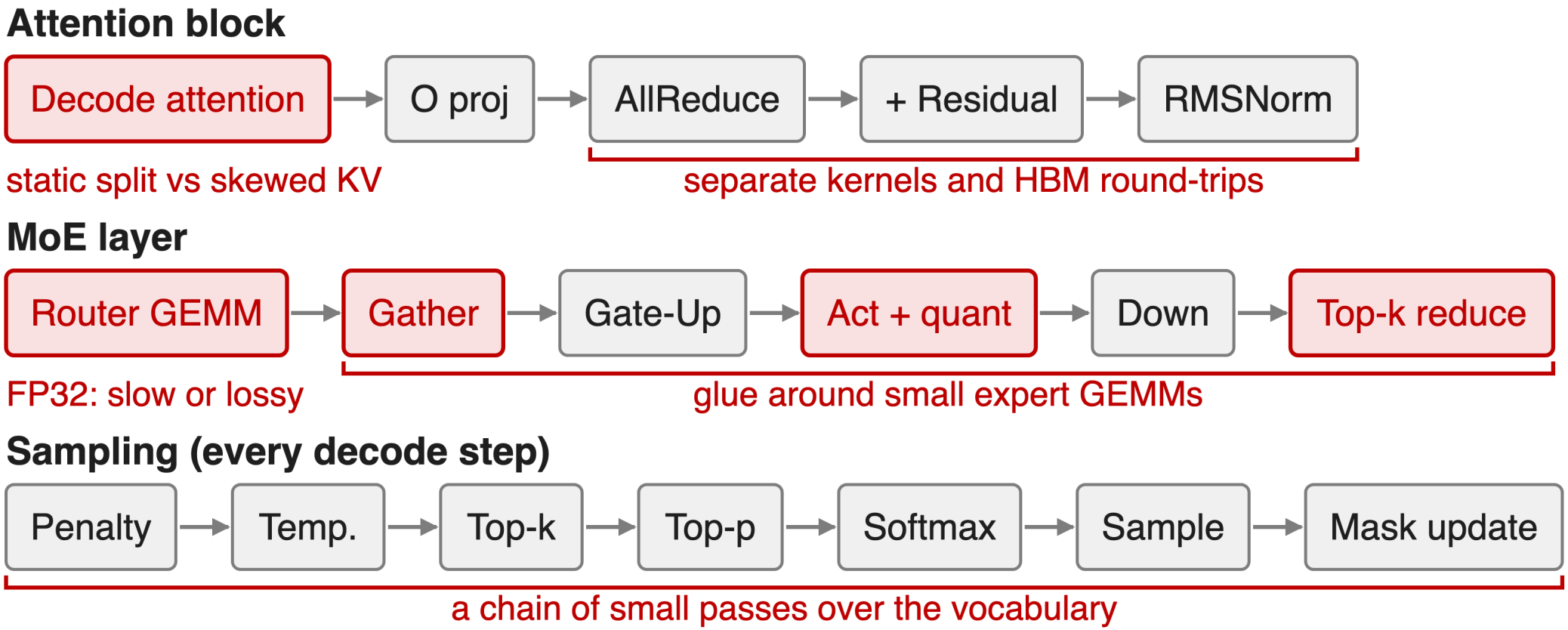
Skewed batches slow decode attention 2.95×

2.95×

slower decode attention under a static split-KV schedule when one 128K request joins 31 × 4K requests

LMSYS / Tencent, "HPC-Ops × SGLang" (Aug 2026), Table 1: H20, FP8 KV cache, 0.186 ms static vs 0.063 ms dynamic

Generic kernels misfit five decode hot spots



Insight: the loss is imbalance and glue, not math

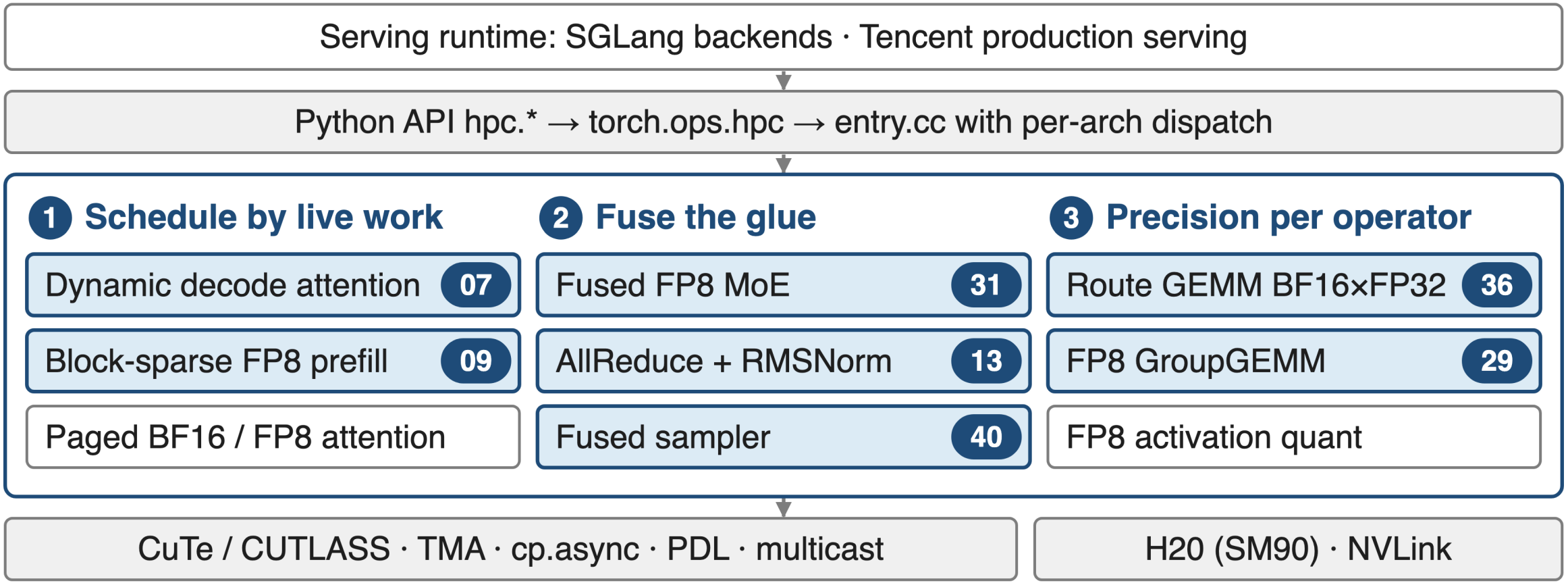
- Uniform 64 × 0.5K batch: static and dynamic schedules tie, **1.00×**
- Skewed 1 × 128K + 31 × 4K batch: same attention math, **2.95×** apart

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Tune each hot path for the production shape distribution, not for one benchmark shape

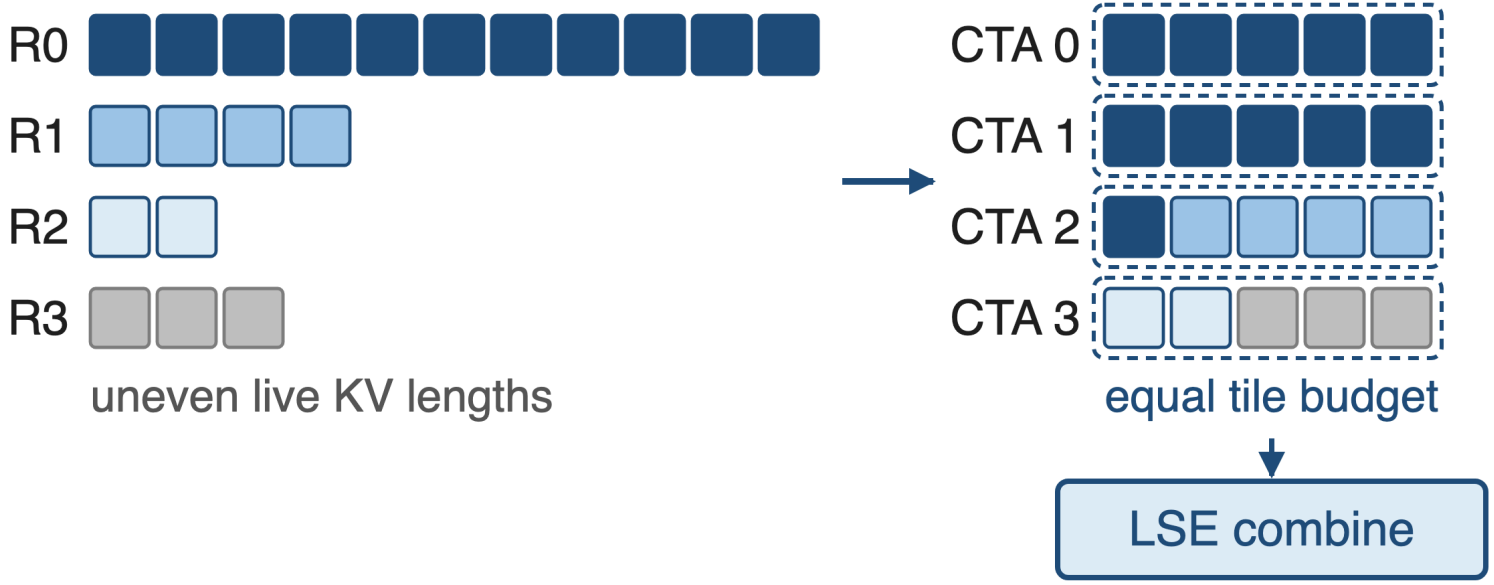
HPC-Ops: one tuned operator per hot path



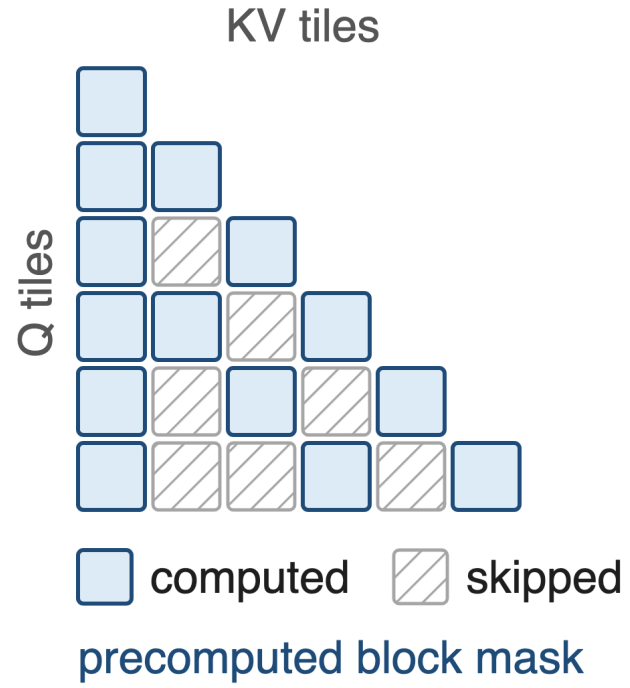
Badges = sibling explainer decks in this series; white chips have no separate deck

❶ Attention schedules by live KV work

Decode: 64-token KV tiles

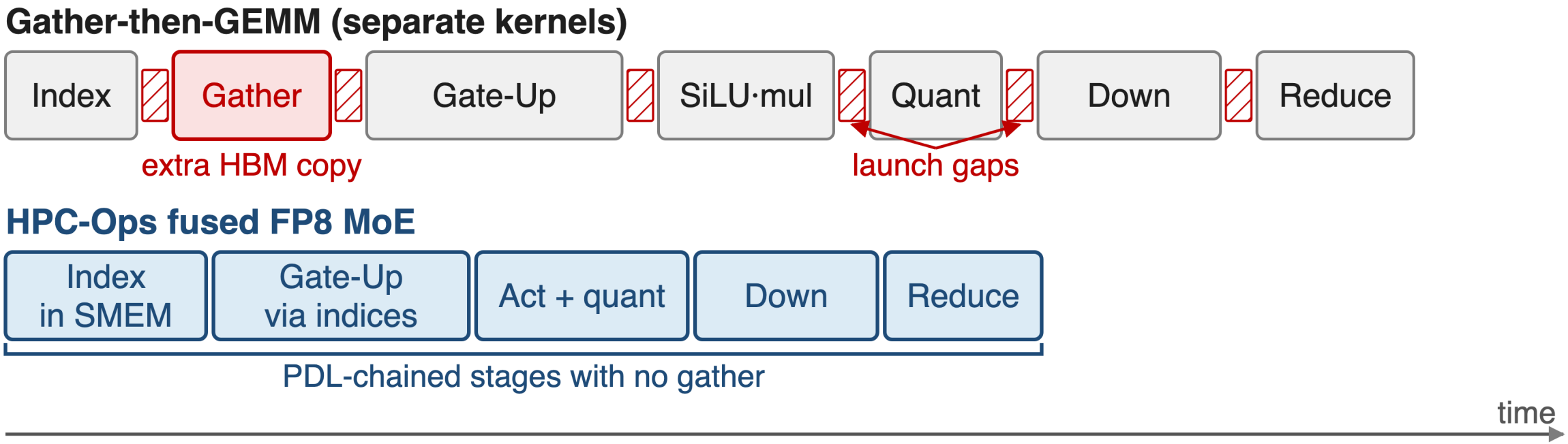


Prefill: FP8 block-sparse



Decode: budget = total 64-token tiles ÷ persistent CTAs (deck 07). Prefill: masked KV tiles never load (deck 09)

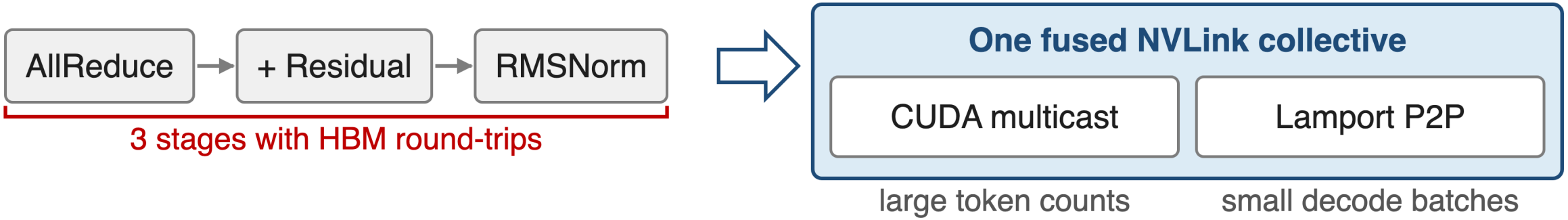
❷ Fused MoE removes the gather and launch gaps



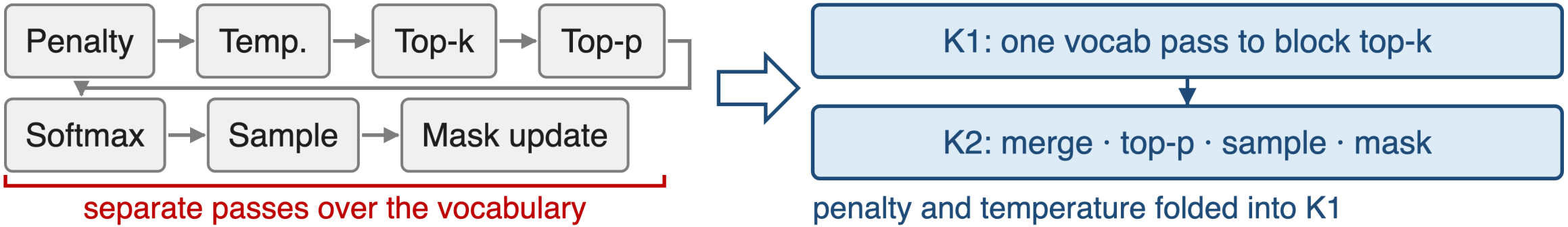
Gate-Up reads tokens through routing indices; PDL overlaps each launch with the previous stage (decks 31, 29)

❷ Fusion shrinks AllReduce and sampler chains

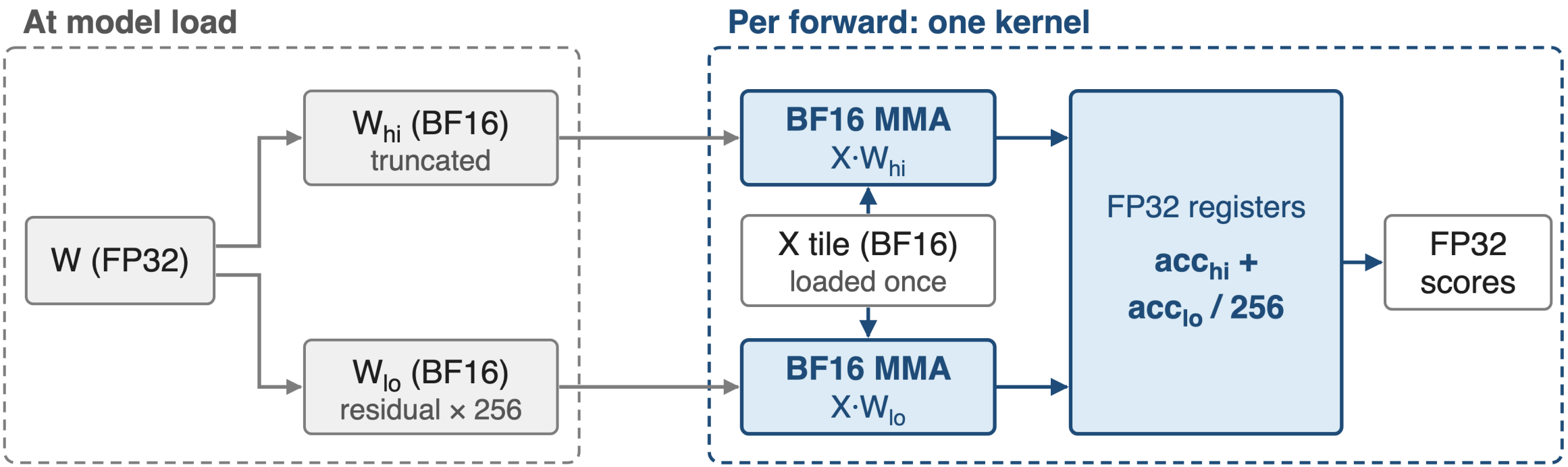
Tensor-parallel output (deck 13)



Sampler (deck 40)

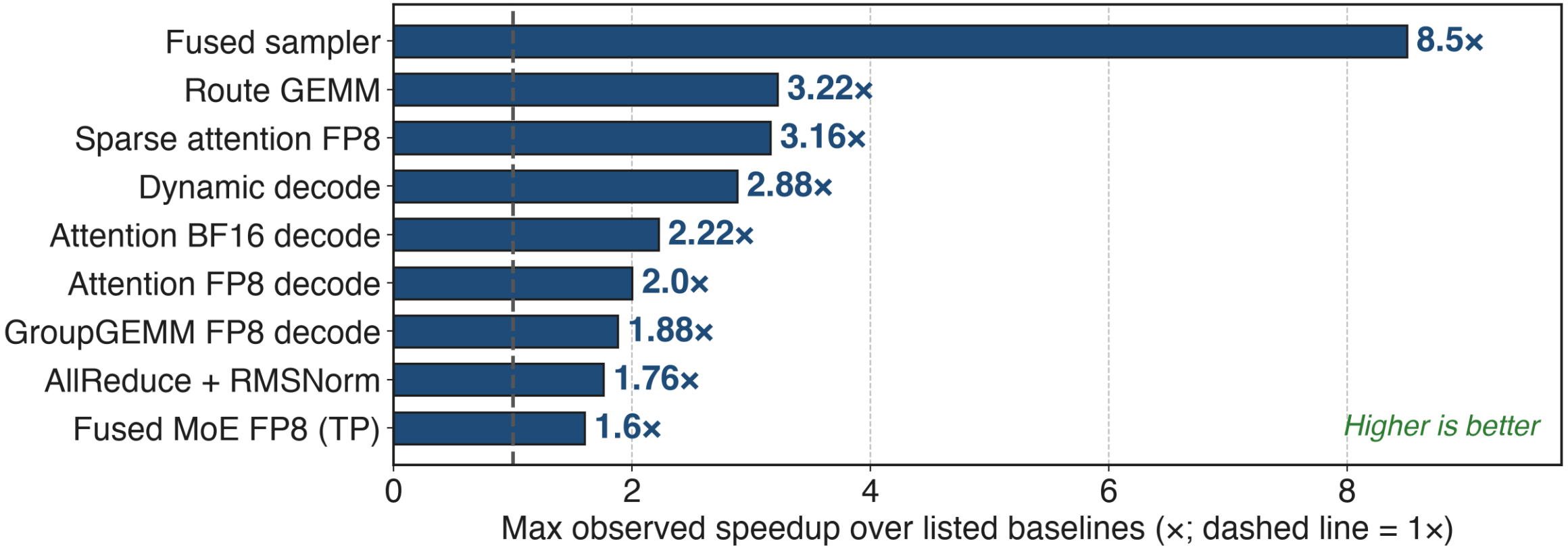


❸ Route GEMM gets FP32 precision from BF16



Max error vs FP32 cuBLAS: 0.00177, versus 0.06464 for TF32 cuBLAS (deck 36)

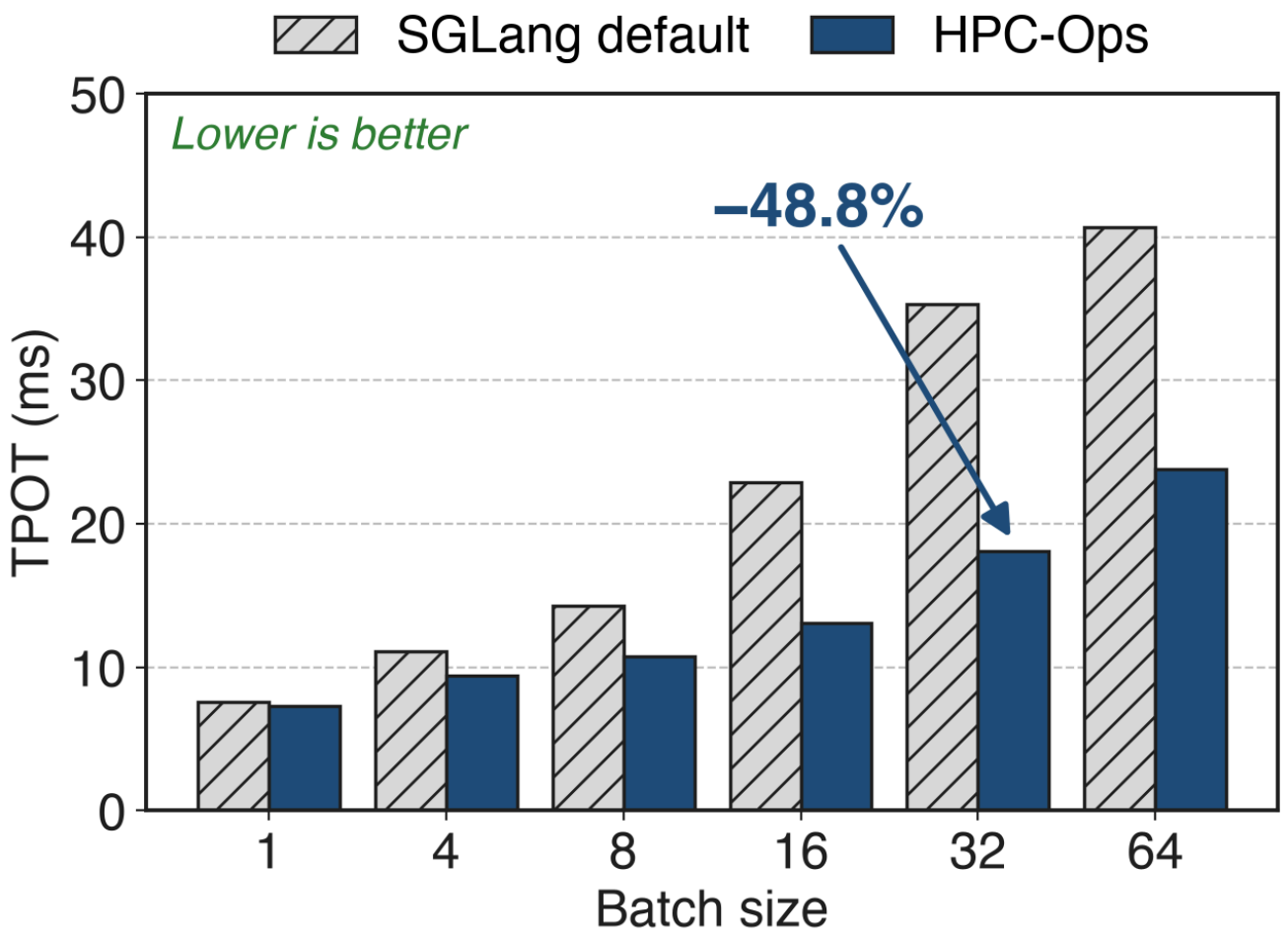
Every hot path gains 1.6–8.5× at its best



README maxima per operator vs. the baselines it lists; results vary by shape

Hy3-FP8 per-token latency drops up to 48.8%

- HPC-Ops Attention + MoE in SGLang, 8 × H20
- TP8, FP8 KV cache, 8K in / 4K out
- TPOT **–15.1% to –48.8%** at batch 4–64



Tencent production gains ~30% QPM

- Tencent serving: **~30%** QPM for Hunyuan, **~17%** for DeepSeek models
- SGLang main: `hpc_ops` attention and MoE backends, Router GEMM auto-dispatch
- H200 check: Hy3-FP8 output throughput **+3.7–5.9%** vs FlashAttention

Gains are H20-tuned and shape-specific

- **Hardware**: tuned for SM90 / H20; Blackwell builds carry few kernels yet
- **Shapes**: head_dim 128 attention; fused AllReduce only for hidden 4096 / 5120 / 7168
- **Not always faster**: MoE **0.98×** at EP8 batch 512; Hy3 MoE on H200 **–4.2%** to +6.3%

Summary

- Production decode shapes are skewed and small: static attention runs **2.95×** slower
- **HPC-Ops**: schedule by live work, fuse the glue, choose precision per operator
- Hy3-FP8 TPOT **–15.1–48.8%** on 8 × H20; **~30%** QPM in Tencent production
- Deep dives: decks 07, 09, 13, 29, 31, 36, 40

https://github.com/Tencent/hpc-ops